

FPT University HCM



Deliverable 1

Topic: Agriculture UAV

**Platform for managing and
coordinating precision agricultural
spraying drone services in the
Mekong Delta region.**

Subject: ITA301

Instructor: *Ngo Dang Ha An*

Ho Chi Minh City, May 2026

PROJECT INFORMATION

1. **Department responsible:** This project was carried out by a group of students currently studying at FPT University Ho Chi Minh City as part of their ITA301 – Systems Analysis and Design course.

List of Project Members

Full Name - Student ID	Function
Le Hoang Khang - SE194654	Project Leader - Business Analyst
Nguyen Anh Hong - SE196203	Research Analyst
Tran Le Phuong Uyen - SE194868	Business Analyst
Nguyen Nhat Chinh - SE196524	System Analyst
Le Nhat Khang - SE196620	Documentation Lead - Business Analyst

2. **Content to be implemented:** Research and analysis of a platform supporting the coordination of precision agricultural spraying drones aims to optimize UAV management, spray scheduling, task progress monitoring, and support the optimization of operational models and management of precision agricultural services.

Research project objectives:

- Optimize human resources, minimize scheduling conflicts and coordination errors.
- Increase the efficiency of pesticide spraying operations using UAVs.
- Monitoring task progress at a visual, operational level and enhancing the transparency of log data.
- Contributing to the advancement of digital transformation in the field of precision agriculture in Vietnam.

3. **Execution time:** May 2026 – August 2026.

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1. Team Assignment

Member	Primary Role	Secondary Role	Main Responsibilities
Le Hoang Khang	Project Leader	Business Analyst	Coordinate projects, analyze requirements, and communicate with stakeholders.
Nguyen Anh Hong	Research Analyst	-	Conduct market research, competitor analysis, data analysis, and find research materials.
Tran Le Phuong Uyen	Business Analyst		Analyze system requirements, identify stakeholders, build business cases, manage project scope, develop SRS and system analysis documents.
Nguyen Nhat Chinh	System Analyst	-	Workflow design, system architecture, and solution analysis.
Le Nhat Khang	Business Analyst	Documentation Lead	Manage documents, prepare reports, and review deliverables.

2. Stakeholders Mapping

2.1 Primary Stakeholders

Stakeholder	Role	Current Pain Point	System Needs	Related Functions	Priority
Administrator(Zhang and al., 2016).	Administering the entire system, managing accounts	Managing users/data with fragmented tools makes it difficult to control	• A centralized tool for managing all accounts. • A tightly controlled	• User Management • Role & Permission Management	High

	, assigning permissions, and monitoring.	access and track errors.	system of delegation of authority. • Check the system logs to look up errors.	• Audit Log / System Logging	
Manager (Kuzmenkov et al., 2026).	Receive requests, coordinate personnel (Operator, Agronomist, Tech) and equipment.	Manual coordination via phone/Zalo easily leads to scheduling conflicts and makes it difficult to track resource status and task progress.	• Monitor the status and availability of personnel and equipment. • Supports efficient task assignment. • Monitor task progress on a centralized platform.	• Schedule & Calendar Management • Mission Assignment / Dispatching • Operations Monitoring Dashboard	High
Farmer (Arsenoaia et al., 2026).	Create service requests, track schedules, receive performance reports, and make payments.	Difficulty in accessing precision spraying services, lack of transparency regarding work progress and post-implementation results.	• Easily schedule services via app/website. • Track your order status visually. • Track invoices and payment status.	• Service Request Management • Order / Mission Tracking • Billing & Invoice Tracking	High
Agronomist (Arsenoaia et al., 2026).	Assess the current state of the crops, and propose	Manually recording survey data and communicating recommendations to the	• Enter the digital survey report directly into the field. • Systematize the list of recommended	• Field Survey Mobile/Web Input • Chemical & Prescription Management	Medium

	appropriate pesticides and dosages.	operations team is slow and prone to information errors.	medications. • Link prescription data to the flight crew.	• Data Sharing Integration	
Drone Operator (Kuzmenkov et al., 2026).	Receive flight schedules, carry out pesticide spraying missions using UAVs, and update on field progress.	There is a lack of a centralized platform for receiving detailed flight plans, tracking equipment status, and updating results.	• Receive mission notifications and flight area maps. • Progress/completion reports are provided directly in the field. • Record the spray log.	• Mission Execution Mobile UI • Spray Log Management • In-field Status Update	High
Technician (Kuzmenkov et al., 2026).	Manage UAV equipment, batteries, charging stations, and perform routine maintenance.	It is difficult to track repair history, equipment wear and tear, and maintenance alerts, which can easily lead to flight disruptions.	• Manage the device and battery lifecycle. • Track maintenance history and costs. • Receive automatic alerts when maintenance is due.	• Maintenance & Repair Management • Drone & Battery Fleet Management • Alert & Notification System	Medium

2.2 Secondary Stakeholders

Stakeholder	Role	Current Pain Point	System Needs	Related Functions	Priority
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Pesticide supplier	Providing product information and ensuring the pharmaceutical supply chain.	It is difficult to accurately predict seasonal and regional demand for pesticides, leading to overstocking or shortages.	• Access to aggregated (anonymized) data on drug consumption. • Update the product catalog on the system.	• Supply Chain Analytics (ReadOnly) • Partner Catalog Management	Low
Drone/UAV supplier	We supply aircraft, batteries, and replacement parts.	Lack of real-world data on hardware failures in the field is crucial for product improvement and optimizing warranty processes.	• Receive statistical reports on common technical issues from the Technician department.	• Hardware Error Reporting & Analytics • Warranty Data Export	Low
Agricultural insurance company	Monitor seasonal risks and offer insurance packages.	Lack of tools to verify the transparency of farming data increases the risk of fraud or errors in compensation.	• Transparently track spraying history and disease outbreak surveys to serve as a basis for risk assessment.	• Historical Data Auditing • Certificate / Log Generation	Medium

Data Analyst / AI Engineer	Collect and analyze data to optimize system algorithms.	Agricultural operational data is scattered and unstructured, making it difficult to train AI models.	• Clean, standardized data sources on area, plant type, dosage, and effectiveness of pest control.	• Data Warehouse / ETL Pipeline • AI Model Integration Interface	Medium
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2.3 Regulatory Stakeholders

Stakeholder	Role	Current Pain Point	System Needs	Related Functions	Priority
Vietnam Ministry of Agriculture and Rural Development	Issue a list of permitted drugs and safety standards.	It is difficult to control service providers from using illegal or banned pesticides, or spraying in amounts exceeding the prescribed limits.	• The system automatically blocks and alerts if there is any prescription of medication outside the permitted list.	• Regulatory Compliance Engine • Banned Chemical Blacklist Filter	High
Vietnam Civil Aviation Authority	Airspace management, flight permits, and drone pilot certification.	Difficulty in monitoring uncontrolled flying devices, risk of collisions, or flying into restricted areas.	• The system must mandate updating no-fly zones maps and verifying the validity of drone operator certifications.	• No-Fly Zone (NFZ) Map Integration • Pilot License Verification	High

Local government	Control pesticide spraying activities and ensure security and order in the area.	Without real-time monitoring tools for large-scale flight operations, response is only provided when incidents or complaints occur.	• Mechanism for automatically generating flight schedule reports or declarations when operating within the area.	• Local Flight Declaration / Reporting • Administrative Export Tool	Medium
Environmental agency	Monitoring the impact of chemicals on water, soil, and air.	It is difficult to assign responsibility and determine the root cause when environmental pollution incidents occur.	• The spraying log (coordinates, time, dosage, wind direction) is stored intact and cannot be edited or deleted.	• Immutable Spray Log Storage • Environmental Impact Auditing	Medium

3. Problem Statement

3.1 Current Issues in Service Coordination

Manually managing drone services via Excel, Zalo, and notebooks creates operational bottlenecks:

- **Long processing times & Information blindness:**The distributed process allows for faster processing times.**17.50 minutes/request**, which reduces the level of satisfaction.**3.44/5.0 (Tanut et al., 2025)**The lack of a centralized platform hinders the ability to share real-time data and makes it difficult for managers to grasp the availability status of resources.
- **Overlapping schedules:**Scheduling conflicts and record-keeping errors amounted to**12.00% (Tanut et al., 2025)**The lack of automatic synchronization mechanisms increases operating costs and reduces the efficiency of resource utilization.

- **Information gaps in expertise:**Data from agricultural experts is fragmented across multiple channels, leading to a loss of management effectiveness.**1–2 hours/day**compile report(**Tanut et al., 2025**)This disruption causes delays in technical communication and inconsistencies in data.
- **Lack of transparency in field data:**Spray logs and operation logs are stored in a fragmented and unstable manner.(**Basir et al., 2024**)This compromises traceability and makes it difficult to verify incidents.
- **Risk of service disruption:**Managing UAV maintenance history and battery lifecycles using manual tools leaves the engineering department in a reactive position, facing the high complexity of battery degradation.(**Alhalabi et al., 2025**)The ANN model combined with the MATLAB RUL Prediction Toolbox achieved a MAPE error of only 2.9% to 3.2% in predicting UAV battery status.

3.2 Stating a Measurable Problem Statement

Operating drone spraying services using manual tools like Excel, Zalo, and mobile phones results in a longer average time for receiving and processing a service request.**17:50**, while the rate of data recording errors and scheduling conflicts reached ,**12.00%**Furthermore, field data is stored in a fragmented, decentralized manner and is difficult to retrieve, while equipment maintenance monitoring is primarily based on experience and manual procedures, increasing the risk of unplanned breakdowns and service disruptions. User satisfaction levels are also only average.**3.44/5.0**This reflects the limitations in the transparency and efficiency of the current operational process.

To provide a quantitative basis for analyzing, evaluating, and specifying system requirements in subsequent stages, the limitations of the manual operating process are modeled into initial state metrics (Manual Baseline) as presented in the table below:

Operating criteria (Metric)	Manual (Baseline)	System Implementation (or System Implementation)	The direct consequences for the DOCX service unit.

Service Request & Booking Time (Tanut et al., 2025)	17.50 minutes / request	4.00 minutes / request	Significantly reduces processing time and eliminates cumbersome manual phone/text confirmation procedures.
Data Errors & Scheduling Conflicts (Tanut et al., 2025)	12.00% data recording errors and calendar conflicts	0.50% (Almost completely eliminates scheduling overlap errors)	The automated scheduling conflict detection system helps eliminate flight scheduling conflicts.
User Satisfaction Score (Tanut et al., 2025)	Average rating: 3.44 / 5.0	Rated 4.78 / 5.0	Improve service quality and provide customers with transparent schedules and logs for pesticide application.
Transparency of field data (Basir et al., 2024).	Discrete, unstable, and distributed storage.	Digitizing and centralizing all Spray Log and Operation Log on the system.	Ensuring transparency, easy traceability, and accurate verification in the event of an incident.
Percentage of equipment serviced on schedule. (Alhalabi et al., 2025).	Manual, passive monitoring based on experience. <i>(based on traditional methods).</i>	The AI model automatically predicts battery degradation with high accuracy (MAPE error of	Preventing unplanned malfunctions avoids unnecessary premature battery replacements, extends the lifespan of the drone, and eliminates the risk of service disruptions during spraying operations.

		only 2.9% – 3.2%).	
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4. Scope of the Project

4.1 In-Scope

The functionalities within the project's scope include the following nine core modules:

1. User and Access Management

- User account management (Registration, login, forgotten password, session management).
- Strict access control based on a model.**Role-Based Access Control (RBAC)**for 6 target groups: Administrator, Manager, Agronomist, Drone Operator, Technician, Farmer.
- **System Logging:**Tracking data impact history and system error lookup is provided exclusively for Administrators.

2. Farmer and Farm Management (Client & Farm Management)

- Manage customer information profiles (Farmer).
- Land parcel and farming area information management: Storing location coordinates (manually/pinned points on the map), area, crop type, and farming history for each production area.

3. Service Request & Prescription Management

- **Received the request:**This allows farmers to create pesticide spraying service requests, select plots of land, and track order status visually through a web/mobile interface.
- **Drug invoice number:**Agronomist support includes updating digitized field survey reports, managing pesticide catalogs, and attaching pesticide prescriptions (types, recommended dosages) directly to each service request.
- **Regulatory Filter:**The system automatically checks and issues a warning if Agronomist's prescription contains an active ingredient on the list of banned/restricted chemicals (meeting the requirements of the Ministry of Agriculture and Rural Development).

4. Resource & License Management

- Manage the drone fleet list (Device code, model, operating status).
- Manage the profile information of Drone Operators (Pilots).
- **Pilot License Verification:**The system stores and tracks the expiration dates of pilots' flight certificates. It will automatically alert or lock

assignment privileges if the flight certificate expires (meeting the requirements of the Civil Aviation Authority).

5. Mission Scheduling and Assignment

- Assist the Manager in planning task execution based on approved prescription and land plot information.
- **Check for resource conflicts:** Displays the busy/available schedules of pilots and equipment. Automatically alerts you to scheduling conflicts or equipment conflicts.
- **Visual coordination support:** Assist the Manager in selecting the appropriate Drone Operator based on the assigned area or pre-defined work location.
- **No-Fly Zone (NFZ) Basic Integration:** Display pre-configured wide-area no-fly zones on the Manager's flight management map to avoid assignment to hazardous areas (meeting the requirements of the Civil Aviation Authority).

6. Mission Tracking

- Real-time mission status monitoring at the business level, excluding drone telemetry data.
- Update the normalized service state transitions: *Pending -> Scheduled -> In Progress -> Completed -> Cancelled*.
- Supports a dedicated, intuitive dashboard for managers to monitor the progress of all flight crews in the field.

7. Spray Log and Flight Log Management

- **Spray Log:** Allows the drone operator to log spraying data after completing a task (actual sprayed area, actual amount of pesticide consumed, wind direction, time).
- **Flight Log:** Record the cumulative flight hours of the UAV and the number of charge/discharge cycles of each battery for depreciation calculation purposes.
- **Data transparency:** The system securely stores log data, automatically synchronizes it, and generates digitized reports that are sent directly to farmers via the system for verification in case of incidents (meeting the needs of the Environmental Agency and farmers).

8. Maintenance Management (Maintenance & Equipment Management)

- Manage the maintenance history, repairs, component replacements, and maintenance costs for each drone/battery device in the Technician department.
- **Automatic maintenance alert:** Automatically send alerts to the technician when the device reaches its operational limit (e.g., UAV reaches 50 flight hours, battery reaches 100 charging cycles) based on synchronized data from the Flight Log.

9. Operational Reporting & Data Export (Reporting & Data Export Dashboard)

- **Internal report:** Provides visual performance indicators (Total number of missions, total area sprayed, amount of pesticide used, projected revenue, pilot performance report).
- **Local Flight Declaration Tool (Export external reports):** Supports the export of flight schedules, flight coordinates, and pilot lists in standard formats for use in declarations, applications for permits from local authorities, or assessments by insurance companies (meeting the needs of local authorities and agricultural insurance companies).
- **Billing & Invoice Tracking (Basic):** Record invoice information, total service fee based on sprayed area, and update payment status. *Awaiting payment / Paid* of Farmer (Excluding complex online payment gateways).

4.2 Out-of-Scope

Content, technologies, and processes that fall outside the scope of this project include:

1. Drone Hardware Development & Control

- Design, manufacture, assemble, or modify drone hardware, batteries, charging stations, or remote-controlled (RC) equipment.
- Developing algorithms for autonomous flight control, automatic takeoff/landing, or real-time obstacle avoidance.
- Remote control of UAVs via the internet eliminates the need for on-site pilot supervision.

2. Advanced Route Optimization & Telemetry (Flight Route Optimization & In-depth Data)

- Automatically scan land plot boundaries using AI or specialized surveying equipment.
- Generate detailed flight paths (Waypoints) for UAVs or optimize aircraft flight paths using specialized algorithms.
- **Real-Time Telemetry Streaming:** The system does not stream detailed real-time technical data from the drone to the server (such as: precise altitude to the second, tilt angle, current wind speed of the aircraft, or livestream video from the drone's camera).

3. Specialized Agriculture Hardware & IoT (Intensive IoT Devices & Sensors)

- Integrating and processing data from multispectral cameras to analyze the National Plant Health Index (NDVI).
- Integrate a high-precision GPS RTK positioning system or manage a ground signal correction base station (RTK Base Station).

- Connect and manage other smart agricultural IoT devices (soil moisture sensors, smart field weather stations).

4. Advanced AI & Image Analytics

- Develop an AI/Machine Learning model to automatically identify pests and diseases through images of land plots.
- Automated drug prescription decision-making or dosage calculation based on analysis of satellite or drone imagery.

5. External Logistics & Heavy Financial Systems

- Supply chain management, physical warehousing, and logistics for delivering pesticides directly to the fields.
- Automatically integrate online payment gateways (E-payment Gateway) with banks or e-wallets (only manual invoice status management is required).
- In-depth business accounting systems, cash flow management, payroll management, or comprehensive agricultural ERP systems.

5. Business Case

5.1 Economic potential of drone-based pesticide spraying services

In the context of developing smart agriculture, numerous experimental studies worldwide have demonstrated that the application of unmanned aerial vehicles (UAVs) brings superior economic efficiency and optimizes production costs.

Research by [Quan et al. \(2023\)](#) Surveys conducted among over 2,000 rice and grain farmers in China have indicated that the application of UAVs in spraying pesticides helps...**increase revenues** significant, **ranging from 434–488 USD/ha**, simultaneously **Save 14.4–15.8 hours** labor per hectare. In parallel, [Li et al. \(2025\)](#) Provide practical evidence from mountainous rice-growing regions regarding **potential to support medication reduction** plant protection inputs **but still maintain and improve production efficiency**. In terms of environmental impact and material costs, [Zanin et al. \(2022\)](#) It is asserted that precision spraying technology has the capability...**Reduce the amount of chemicals used while still ensuring effective pest control**. thereby limiting negative impacts on agricultural ecosystems.

However, the economic and technical evidence presented above only reflects the independent operating performance of **UAV hardware technology itself** As the scale and frequency of service provision expand commercially in cooperatives or large enterprises, complex information management problems will arise. Without a **specialized information system tools** To optimize resource allocation, businesses risk undermining the inherent economic advantages of

UAV technology due to soaring logistics costs, pilot scheduling conflicts, and disruptions in agricultural expertise.

Research gap: The current literature primarily focuses on the technical features or biological effectiveness of hardware drones. Conversely, research into designing a specialized Farm Management Information System (FMIS) platform that coordinates resources, monitors operational progress, and manages the spraying/flight log chain remains a significant gap. This academic motivation is the driving force behind the authors' proposed design solution for this project.

To address this research gap, the project focuses on answering three research questions: **Research Questions - RQs**) serves as a guiding principle throughout:

- **RQ1:** What operational bottlenecks and resource management risks exist in the manual UAV spraying service delivery model?
- **RQ2:** What core requirements should a UAV service management and coordination information platform meet?
- **RQ3:** How does the proposed system modeling and software architecture help improve the operational performance indicators (KPIs) of the service unit?

5.2 Limitations of the current management model

Although drone technology offers many practical benefits, much of the service management and coordination process in Vietnam is currently still carried out using rudimentary, fragmented tools (Excel, Zalo, phone calls). This leads to core information system bottlenecks including:

- **Resource information asymmetry:** Management cannot visually monitor the real-time operational readiness status (available, busy, under maintenance) of drones, batteries, and field pilots to optimize assignments.
- **Scheduling Conflicts:** Manual procedures are prone to errors, overlapping pilot duty schedules, or duplication of flight equipment during peak seasons.
- **Disruption of specialized data flow:** Recommendations regarding the type and dosage of pesticides from agricultural engineers (Agronomists) are manually communicated to pilots and are prone to delays or inaccuracies when loading the aircraft's operating configuration.
- **Lack of transparency and data traceability:** The lack of automated field data recording tools results in a data deficit in the system. *Spray Log* (Spray log) and *Flight Log* (Flight logs) provide transparent retrieval and reporting for customers and regulatory authorities.
- **Passive technical risks:** Managing UAV maintenance history and battery lifecycles using rudimentary tools makes the technical department less

proactive, increasing the risk of unexpected breakdowns that disrupt field services.

5.3 Expected Value of the Management and Coordination Platform

Unlike the direct economic benefits from drone hardware, the proposed management and coordination platform focuses on: **Digitizing processes, optimizing resource allocation, and supporting enhanced data traceability.** Based on theoretical models of agricultural information systems, digitizing processes helps minimize work processing time and reduce operational errors to the greatest extent possible.

To ensure consistency between the theoretical basis, the real-world problem, and the software solution, the development team... **Evidence Traceability Matrix** below:

Literature Evidence	Operational Problem	Proposed Function	Evaluation Metric
Optimizing flight scheduling is a key factor in performance. (Kharchenko et al., 2022).	Manual coordination via Excel/Zalo can easily lead to scheduling conflicts between pilots and drones.	Module 5: Mission Scheduling & Resource Conflict Detection	Scheduling time & Schedule conflict rate
Information system automation helps eliminate management lag. (Paraforos et al., 2018).	It takes a lot of time at the end of the day to compile performance reports.	Module 9: Operational Reporting & Automated Data Export	Time to generate operational reports

The ability of UAVs to track large amounts of data is essential.(Fernández-Caramés et al., 2019).	Field data is scattered, and there is a lack of tools for storing Spray Logs/Flight Logs.	Module 7: Flight Log & Append-Only Spray Log Management	The percentage of tasks with complete and transparent logs.
Fleet reliability depends on maintenance policies.(Kharchenko et al., 2022).	Passive maintenance management easily leads to missed schedules and flight disruptions.	Module 8: Maintenance & Fleet Life-cycle Management	Percentage of equipment serviced on time

The KPIs below are proposed as a hypothetical scenario for the purpose of analyzing the research scenario of the project.(**Assumption for Project Scenario Analysis**), serves as a baseline and a target for measuring design effectiveness at the end of the project.

Evaluation criteria (Metric)	Manual (Baseline)	Design objectives (Target)	Source
Service request processing time	15–30 minutes / per request	Under 5 minutes / request	Project Assumption (Based on a Cloud-based centralized processing mechanism)(Itkin et al., 2016).
Rate of scheduling errors or overlaps	10–15%	Less than 2%	Project assumption (Based on automatic schedule conflict blocking logic)(Kharchenko et al., 2022)).
Timeframe for compiling	It takes 1–2 hours/day	Under 10 minutes/report	Reference to FMIS information system

operational reports.			research (Paraforos et al., 2018).
Transparency of field data	Unstable, distributed storage	Approximately 95% of missions have both Spray Log and Flight Log.	Reference model for secure UAV data storage. (Fernández-Cara més et al., 2019).
Percentage of equipment serviced on schedule.	Not yet under focused monitoring	At least 95% of the equipment is maintained on schedule.	Project Assumption (Based on flight limit threshold alert trigger) (Kharchenko et al., 2022).

6. Work Breakdown Structure (WBS)

6.1 Phase 1 – Initiation & Business Analysis (Week 1–3)

- **6.1.1 Project Initiation**

- Form a project team.
- Choosing the topic: Agriculture UAV Service Management System
- Define project objectives
- Define the project scope.
- Identify the main Deliverables

- **6.1.2 Stakeholder Analysis**

- Identify Primary Stakeholders
- Identify Secondary Stakeholders
- Identify Regulatory Stakeholders
- Stakeholder Needs Analysis
- Building Stakeholder Mapping

- **6.1.3 Business Analysis**

- Analysis of the current operational process (AS-IS Process)
- Identify the pain points of the parties involved in manual coordination.
- Identify Opportunities for Improvement
- Construct a Problem Statement (Measurable)
- Define Business Objectives
- Building Scope Definition

- **6.1.4 Project Planning**

- Develop a Business Case and a Key Performance Indicator (KPI) framework.

- Cost-Benefit Analysis
- Construct a Work Breakdown Structure (WBS).
- Create a Project Timeline
- Identifying Milestones
- Build a Risk Register
- Develop a Communication Plan

- **Deliverable: D1 – Project Proposal**

6.2 Phase 2 – Requirements Engineering (Week 4)

● **6.2.1 Requirement Elicitation**

- Gather requirements from Business Stakeholders (Farmer, Manager, Operator, Agronomist, Technician)
- Gather requests from Regulatory Stakeholders (Department of Agriculture, Civil Aviation Authority, Local Government)
- Define Functional Requirements (FR) based on 9 core modules.
- Identify Non-Functional Requirements (NFR)
- Define Business Rules (Prescription approval process, flight scheduling rules)
- Check Regulatory Constraints (Publicly Prohibited Substances Filter, Pilot License Expiration Date)

● **6.2.2 Requirement Analysis**

- Identify System Actors (6 groups of system users)
- Build User Stories for each functional subsystem.
- Establish Acceptance Criteria
- Design the overall and subsystem Use Case Diagram.
- Detailed description of the main Use Cases (Appointment scheduling, Prescribing medication, Assigning tasks, Updating logs)

● **6.2.3 Requirement Documentation**

- Xây dựng Software Requirements Specification (SRS)
- Review requirements with stakeholders.
- Check the completeness and consistency of the Requirements
- Approve the request documents.

- **Deliverable: D2 – Software Requirements Specification (SRS)**

6.3 Phase 3 – System Modeling (Week 5–6)

● **6.3.1 Process Modeling**

- Designing a Context Diagram
- DFD Design Level 0
- Design Level 1 DFDs for the modules (Service Requests, Assignments, Log Management, System Logs)
- Designing a seamless workflow from Prescription (Agronomist) to Flight Scheduling (Manager)

- Check the balance of the DFD.
- Business Process Review (TO-BE Process)
- **6.3.2 Object Modeling**
 - Complete the Use Case Diagram
 - Designing a Class Diagram for a Resource and Service Management System
 - Design Sequence Diagrams for Key Business Flows
 - The Activity Diagram design describes the logic of task state transitions (Pending $\rightarrow$ Scheduled $\rightarrow$ In Progress $\rightarrow$ Completed $\rightarrow$ Cancelled)
- **6.3.3 Data Modeling**
 - Conceptual Data Model Design
 - Design an Entity Relationship Diagram (ERD) for the system (including entities: Account, Land Plot, Prescription, Flight Schedule, Equipment/Battery)
 - Designing Data Models for Flight Logs and Spray Logs
 - Designing a storage structure for the List of Prohibited Substances and the Map of Widespread No-Flight Zones (NFZs).
 - Third Normal Form (3NF)
 - Build a detailed Data Dictionary.
- **6.3.4 Traceability Management**
 - Xây dựng Requirements Traceability Matrix (RTM)
 - Mapping Requirements với Use Cases
 - Mapping Requirements với Data Model
 - Mapping Requirements với Design Components
- **Deliverable: D3 – System Modeling Package**

6.4 Phase 4 – System Design (Week 7–9)

- **6.4.1 UI/UX Design (Week 7)**
 - Create User Personas and User Journeys for specific Actors (Dispatch Manager, Field Operator).
 - Design Wireframes & UI Prototype for Web System (Admin/Manager/Dashboard Monitoring)
 - Mobile UI Layout/Prototype design for Agronomist (record entry), Drone Operator (progress reporting/log entry), and Farmer (scheduling/order tracking).
 - Design Navigation Flow between business screens
- **6.4.2 Database Design (Week 7)**
 - Physical Data Model Design
 - Database Schema Design
 - Designing a mechanism to ensure data integrity and traceability for Spray Log and Flight Log data.

- Designing Relationships and Data Constraints
- Check data referential integrity
- **6.4.3 Architecture Design (Week 8)**
 - Thiết kế System Architecture (Logical Architecture & Physical Architecture)
 - Choose a Technology Stack (suitable for Web Apps and Responsive/Mobile Web interfaces)
 - Designing an integrated digital map display solution (NFZ Map Basic Integration on the dispatch screen)
 - Deployment Topology Design
- **6.4.4 Security & Quality Design (Week 9)**
 - Designing Authentication Models & Managing Sessions (Session/Token Management)
 - Design a detailed Role-Based Access Control (RBAC) model for 6 Actors (Admin, Manager, Agronomist, Operator, Technician, Farmer)
 - Analyze security threats and design data protection solutions, including system logging for administrators.
 - Develop a Quality Assurance Plan
 - Thiết kế Test Strategy (Unit Test, Integration Test, User Acceptance Test)
 - Mapping Test Cases với Use Cases
 - Perform a trade-off analysis between different software architecture options.
 - Develop an Operations & Maintenance Plan.
- **Deliverable: D4 – System Design Document**

6.5 Phase 5 – Final Evaluation & Defense (Week 10)

- **6.5.1 Final Preparation**
 - Complete the UI/UX Prototype Demo (connecting the main business flow from Web to Mobile layout).
 - Summary of all delivered products (Deliverables D1, D2, D3, D4)
 - Check for cross-consistent data between SRS, UML diagrams, and the System Design Document.
 - Verify the completeness of the traceability matrix (RTM)
 - Finalize the structure and formatting of the project document.
- **6.5.2 Presentation & Defense**
 - Prepare presentation slides for your project report.
 - Implement Mock Defense (Internal team test protection)
 - Anticipate and prepare a set of Q&A questions for the defense session (especially questions about the drug detection mechanism,

the authorization of the 6 actors, and the solution for handling data logs).

- Finalize the summary report.
- Submit the final product and documentation to the Board.
- **Deliverable:** D5 – Final Presentation & Defense

7. Evidence Traceability Matrix (ETM)

7.1 Purpose

The Evidence Traceability Matrix (ETM) is designed to ensure traceability between stakeholder needs, business requirements, functional requirements, and the system's analysis and design products.

ETM helps the project team:

- Ensure that all stakeholder needs are reflected in the system.
- Verify the completeness and consistency of the requirements.
- Demonstrate that each requirement is supported by evidence in the models and design documents.
- Assist in project scope assessment and requirements change management.
- Increase the transparency and verifiability of delivered products.

7.2 Evidence Traceability Matrix

Stakeholder needs	Business requirements	Functional requirements	Related subsystem	Proof (Evidence)
Managers need to avoid scheduling conflicts between pilots and drone operators.	Optimizing resource allocation	Calendar conflict detection system	Mission Scheduling & Assignment	Use Case Diagram, Activity Diagram
Managers need to monitor resource status.	Enhance monitoring capabilities.	Display the available/busy status of drones and personnel.	Resource Management	Dashboard, Class Diagram
Farmers need to monitor the service progress.	Increase transparency	Track order and task status	Service Request & Mission Tracking	Use Case Diagram, Sequence Diagram

The farmer needs to see the service invoice.	Transparent payments	Track invoices and payment status	Billing & Invoice Tracking	UI Prototype, Data Model
Agronomists need to manage electronic prescriptions.	Standardize the survey process.	Create and manage digital prescriptions.	Prescription Management	Use Case Diagram, Class Diagram
The Ministry of Agriculture and Rural Development requires control over banned chemicals.	Ensure compliance with regulations.	Check for substances on the banned list.	Regulatory Filter	Business Rules, Chemical Blacklist
The Civil Aviation Authority requires verification of pilot certificates.	Ensure compliance with flight regulations.	Verify the validity of the pilot's certificate.	Resource & License Management	Sequence Diagram, ERD
Managers need to monitor task progress.	Operational control	Receive real-time mission status updates.	Mission Tracking	Dashboard, Activity Diagram
Drone operators need to keep a log of pesticide spraying.	Standardize operational data.	Create and save a Spray Log	Spray Log Management	Use Case Specification, ERD
Technicians need to track maintenance history.	Improve device reliability	Manage repair and maintenance history	Maintenance Management	Class Diagram, ERD
Technician needs automated maintenance alerts.	Reduce the risk of malfunctions.	Automatically generate maintenance notifications.	Maintenance Management	Business Rules, Activity Diagram
Environmental agencies need to access spraying data.	Ensuring data transparency	Store the Spray Log and Flight Log in full.	Log Management	Audit Design, Database Design

Administrators need to manage accounts centrally.	Strengthening system governance	User management and permissions	User & Access Management	RBAC Model, Use Case Diagram
The administrator needs to monitor system activity.	Ensuring information security	Record and retrieve System Log	System Logging	Audit Log Design, Class Diagram

7.3 Assessing the level of coverage required

The tracing results show that all the needs of the primary and managerial stakeholders have been mapped to their respective functional requirements and supported by the system's analysis and design models. This ensures that the platform for managing and coordinating agricultural drone spraying services fully meets the business objectives, management requirements, and legal compliance requirements identified within the project scope.

8. Timeline

Week	Phase	Main Activities	Deliverables / Milestones
Week 1	P1 – Project Initiation	The project kickoff involves forming a team, selecting a topic for UAV Service Management, and defining initial goals and scope.	Group list + Topic Proposal
Week 2	P1 – Business Analysis	Conduct a survey of the current process (AS-IS), analyze manual coordination bottlenecks, and build Stakeholder Mapping (6 target groups).	Preliminary draft of Stakeholder Map + Business Case
Week 3	P1 – Project Planning	Complete the Business Case (target KPIs), build the Work Breakdown Structure (WBS), Project	M1 – Milestone 1: Proposal Approval D1 — Project Proposal

		Timeline, Risk Register, and review the Proposal.	
Week 4	P2 – Requirements Engineering	Gathering requirements (elicitation), analyzing functional/non-functional requirements, building user stories, designing use cases, and defining business rules and regulatory constraints.	M2 – Milestone 2: Baseline Requirements D2 — Software Requirements Specification (SRS)
Week 5	P3 – System Modeling	Design Context Diagram, DFD Level 0 & 1 (Process Modeling), build the interconnected prescription flow $\rightarrow$ flight schedule, and sketch Class & Sequence Diagram.	Data Flow Diagram (DFD) + UML Draft + Schedule 1 (PT1)
Week 6	P3 – Modeling Completion	Complete the entire UML Diagram (Class, Sequence, Activity, task state), design the ERD (Data Modeling), build the Data Dictionary and the RTM trace matrix.	M3 – Milestone 3: Finalizing the system model D3 — System Modeling Package
Week 7	P4 – UI & Data Design	Build User Personas/Journeys, design Wireframes & UI Prototypes (Web Admin/Manager and Mobile interface for Operator/Agronomist/Farmer), design Physical Data Models.	UI Wireframes/Prototypes + Physical Data Model
Week 8	P4 – Architecture Design	System architecture design (Logical/Physical), technology stack selection (Web/Responsive), design of digital map integration solution (NFZ Map Basic Integration), and deployment topology.	System Architecture Document + Schedule 2 (PT2)
Week 9	P4 – Security &	Perform Trade-off Analysis, design RBAC (6 Actors)	M4 – Milestone 4: Design Completed

	QA	authorization system, establish data integrity mechanisms for logs, and develop a QA Plan, Test Strategy, and Operations Plan.	D4 — System Design Document
Week 10	P5 – Final Evaluation & Defense	Review system consistency, finalize UI/UX Prototype Demo (Web-Mobile workflow connection), run internal Mock Defense, prepare slides and Q&A questions for review.	M5 – Milestone 5: Project Acceptance D5 — Final Presentation & Defense

9. Milestone

Milestone	Week	Deliverable
Project Proposal Completion	Week 3	D1 – Project Proposal
Requirements Specification Completion	Week 4	D2 – Software Requirements Specification (SRS)
System Modeling Completion	Week 6	D3 – System Modeling Package
System Design Completion	Week 9	D4 – System Design Document (SDD)
Final Presentation & Defense	Week 10	D5 – Final Presentation & Defense

10. Risk Register

ID	Risk	Description	Impact	Mitigation
R1	Scope Creep	The project generated additional requirements or changes to the scope of features during implementation.	This increases the workload, affecting the progress and quality of deliverables.	Closely manage the Requirements Traceability Matrix (RTM); perform Trade-off Analysis in Phase 4 when requirements change.
R2	Incomplete /Incorrect Requirements	The system requirements were collected incompletely or inaccurately compared to actual business requirements.	This leads to incorrect modeling (DFD, UML) and database design, requiring rework.	Gather requirements from all three stakeholder groups (Primary, Secondary, Regulatory); thoroughly review the SRS document before packaging D2.
R3	Team Member Delay	Team members failed to complete the decomposition tasks in the WBS on time.	This has slowed down the overall progress of the entire project, affecting the Milestone milestones.	Clearly assign roles to each small task in the WBS; hold regular weekly team meetings to check progress (Communication Plan).
R4	Incorrect Time Estimation	The estimated time for performing design and modeling tasks was inaccurate.	This caused delays in product delivery milestones (D1–D5) and progress reporting milestones (PT1, PT2).	Apply a fixed timebox for each phase, closely following the 10-week timeline; prioritize completing core business processes first.
R5	Low accuracy GPS/Coordinates	Location data discrepancies occur when the Farmer selects a plot of land or when tracking flight paths.	This affects the determination of the actual spray area and the accuracy of the Spray Log.	The database structure design supports high-precision coordinate formatting; the design includes features that allow adjusting/re-pinning the center of land

				parcels on the digital map.
R6	Data connection lost at the scene.	Drone operators or agronomists who do not have a stable internet connection in the field to update logs.	Failure to update mission status in a timely manner risks losing or disrupting flight log data.	Mobile interface design supports temporary storage in local memory (Local Cache/Web Storage) and automatic synchronization (Data Sync) upon reconnection.
R7	Weather affects operations .	Unexpected bad weather (heavy rain, strong winds) disrupted scheduled flights.	This causes duplication and congestion in equipment/pilot scheduling when flights are backed up.	The flexible task state transition logic design (Activity Diagram) allows managers to perform "Re-schedule" (quickly change schedules) directly on the dispatch screen.
R8	No-fly zone violations & Prohibited substances	Drones flying into restricted airspace (military, airport) or spraying prohibited chemicals by mistake.	Serious legal risks for businesses; the system was denied approval by Regulatory Stakeholders.	It integrates NFZ Map Basic Integration digital map display and a Banned Chemical Blacklist data structure design to automatically block/warn customers right from the order creation and scheduling stage.
R9	Information Security & Data Leaks	The system was accessed without authorization, or the spray log data was maliciously altered.	Data insecurity leads to reduced data transparency for customers (Farmers).	Design a robust RBAC authorization model for 6 Actors; apply a data integrity and auditability design to the logging system.
R10	Increase the complexity of the design.	The challenge lies in modeling the interconnected data flow from prescriptions to	Increased design time, and cluttered Level 1 DFD diagrams and Class Diagrams.	The system architecture is designed with a clear modular approach (Logical Architecture); separating the

		flight schedules and maps.		resource management subsystem from the map monitoring subsystem.
R11	Errors in calculating area/chemicals	Incorrect calculations of pesticide dosage or spray area lead to inaccurate end-of-day performance reports.	Failure to achieve the quantitative targets set out in Section 3.2; damage to the credibility of the management platform.	Perform simulation test cases on the algorithm for calculating area based on land plot coordinates; thoroughly review the data dictionary.
R12	Deliverable Delay	The delivered products (D1–D5) were not packaged and standardized in the correct format on time.	This directly impacts the scores awarded for progress milestones in the course.	Set internal completion deadlines 2–3 days before the official Milestone for the course to allow time for cross-checking for consistency.

11. Conclusion

Project **"A platform for managing and coordinating precision agricultural spraying drone services"** built for **Optimization and reduction** The system addresses bottlenecks in manual operational processes at current agricultural service units and cooperatives. Instead of fragmented management via Excel, Zalo, or phone, which easily leads to scheduling conflicts, lack of information on field resources, and disruptions in the flow of specialized information, the system aims for comprehensive digitalization of the service reception and processing procedures.

Through nine core functional modules clearly defined within the deployment scope, the platform ensures the achievement of realistic quantitative goals, including:

- Optimize service request processing speed to under 5 minutes.
- Improve the accuracy of pilot scheduling and assignment, reducing the error rate to below 2%.
- Ensure transparency of field data by aiming for a spray log and flight log storage rate of approximately 95% to facilitate cross-retrieval.
- Automate report generation and quickly export external declaration data.

Besides solving the problem of optimizing internal operations, the system also plays a crucial role in controlling legal compliance regarding flight safety and the list of prohibited chemicals as regulated by relevant authorities. The close coordination between the requirements tracing matrix (RTM), system modeling (DFD, UML), and the design of the access control architecture in subsequent phases will be a solid foundation for the team to realize the software solution, promoting sustainable digital transformation in the precision agriculture services sector in Vietnam.

12. AI Audit Log

The log records the interaction between the team and the AI tools (Claude, ChatGPT, Gemini) during Phase 1 – Problem & Business Analysis (W1–W3), according to two DTC components: Decomposition and Pattern Recognition. Each entry shows a full contextual prompt, a summary of the AI output, the acceptance/modification/rejection decision with reasons, and the corresponding verification step.

STT	Week / Phase	DTC Component	AI Tool	Prompt	AI Output	Decision & Refinement	Verification
1	W1 / Phase 1 — Problem & Business	Decomposition	ChatGPT	For the project "Platform for managing and coordinating precision agricultural spraying drone services in the Mekong Delta" serving cooperatives and spraying service units, currently managed manually via Excel/Zalo/phone, please help break down the operational problem into main problem groups (business problem decomposition) to determine the scope of functional modules.	AI proposes four problem groups: (1) scheduling & resource coordination, (2) operational log recording, (3) legal/flight safety compliance, (4) performance reporting & analysis.	Accept 4 groups as the framework, but separate group (2) into 2 separate modules (Spray Log Management and Maintenance Management) because of different actors and data flows (Drone Operator vs Technician); add the User & Access Management group that AI missed → a total of 9 official modules (Section 5) instead of the original 4 combined groups.	Compare this with the list of 6 identified Actors (Section 4), ensuring that each module is associated with at least one specific actor.

2	W1 / Phase 1 — Problem & Business	Pattern	Gemini	This study summarizes common design patterns/practices in academic research from 2019-2026 on agricultural UAV monitoring and management systems, particularly concerning data integrity and traceability, as a reference for the project's Business Goals.	AI listed 12 articles with common patterns: blockchain-based traceability, FAIR data principles, real-time monitoring dashboard, cloud-based fleet management.	Accepted 10/12 sources because they closely adhere to the agricultural UAV domain; rejected 2 sources on urban logistics UAVs due to domain deviation. Decided NOT to apply the "blockchain-based traceability" pattern to the project scope due to resource/time constraints; replaced it with the simpler Data Integrity & Auditability principle (RBAC + audit log) in Section 9.3 and risk R9.	Verify that each DOI link provided by AI exists and contains correct content before including it in Section 13 References.
3	W2 / Phase 1 — Problem & Business	Decomposition	Claude	With the six identified stakeholder groups (Farmer, Drone Operator, Manager, Technician, Regulatory Stakeholders, Administrator), break down the needs of each group into specific goals and corresponding solutions (need-goal-solution mapping) to serve the Requirement Traceability Matrix.	The AI generated a mapping table for 6 stakeholders, each with 1-2 needs, totaling 9 rows.	Accepting the mapping framework but modifying the Regulatory Stakeholders' needs — the AI generally stated "needs transparent data," but the team specified it as "needs access to pesticide application data" linked to actual Log Management. Adding 3 Manager needs (real-time progress monitoring) that the AI missed → the RTM table now has 10 rows instead of 9.	Cross-check the RTM table with the Data Dictionary — each solution must point to the correct entity/table that has been defined.

4	W2 / Phase 1 — Problem & Business	Pattern	ChatGPT	List 10 typical risks for a software project managing the operation of IoT/drone devices in Vietnam, considering technical, legal, and operational aspects.	AI lists 10 common risks (scope creep, API integration, data security, regulatory changes, etc.), including the requirement to "comply with FAA Part 107".	Hallucination: FAA Part 107 is a US aviation regulation and does NOT apply in Vietnam — it is replaced by domestic regulations (Decree on the management of unmanned aircraft, CAAV regulations). Accepting 7 out of 10 general risks, excluding 3 irrelevant risks (e.g., third-party payment gateway risk because D1 does not yet have a payment module), adding 5 specific agricultural/drone risks → completing 12 risks R1-R12 (Section 10).	Unable to verify the legal requirements themselves, the risk flagging group R8 indicated "further review with experts/instructors is needed" rather than a definitive statement.
5	W3 / Phase 1 — Problem & Business	Decomposition	Claude	Help restructure the Problem Statement and Business Goals content according to a standard layout (current state – pain points – desired state – measurable goals), based on the draft problem description shared via Excel/Zalo/team scheduling.	AI proposed a clear four-part structure and rewrote it with four quantifiable sample objectives (reduce processing time by 50%, reduce scheduling errors by 90%...).	Hallucination: Four AI-generated, baseless target figures. Accept the four-part structure but replace all the figures with team-agreed metrics based on the project's actual scope (processing time under 5 minutes, schedule errors under 2%... — Section 11).	Compare the final four quantitative objectives with the system's actual measurability (which logs/modules record that data) before including them in the Conclusion.

6	W3 / Phase 1 — Problem & Business	Pattern	Gemini	Comparing the advantages and disadvantages of manual operational management via Excel/Zalo versus a digital platform with an audit trail, in the context of medium-sized agricultural cooperatives in the Mekong Delta, serves as the argument for the rationale for digital transformation in Business Goals.	The AI presented a comparison on chart heavily favoring the advantages of digital platforms, without mentioning the costs or barriers to implementation.	Critical Thinking: The team recognized AI's bias towards technology, overlooking the switching costs and digital skills barriers faced by rural users. They added a more balanced perspective—acknowledging the challenges of training Farmer/Drone Operators in the limitations section instead of simply praising the solution.	Instead of directly using AI-generated text, the team rewrote the argument from a more objective perspective, based on local observations.
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